

EE 4113 – ECE Lab 4
Spring 2026
Lab #04 Report

Student 1 Name: Ivan Garcia
UTSA ID: eaa108
Section: 0B1
Group Number: 19

Student 2 Name: Tristan Herring
UTSA ID: xaw089
Section: 0B1
Group Number: 19

Objective:

In this lab exercise, we explore the non-ideal behavior of op-amp circuits and explore how the circuit's operating bandwidth and gain are interdependent. Sometimes when using op-amps in high-gain amplification circuits, it is desirable to have a fixed gain across an operating range of frequencies (i.e., bandwidth), but this is not the case. The maximum gain that the op-amp circuit can operate at is limited, based on the bandwidth of the signal that the circuit intends to amplify. In poorly designed circuits, higher frequency components in a signal may not experience the same gain as lower frequency components. This results in a corrupted and/or distorted signal at the amplified output. This phenomenon is due to the Gain-Bandwidth-Product or GBP. In this exercise, we will find that by using cascaded op-amp circuits, we may relieve the constraints that the GBP has on bandwidth and gain. The circuit that we use to demonstrate the GBP principle is an ADC signal conditioning circuit, where a small input signal (centered around 0V) is provided with a DC offset, then amplified.

Equipment List:

Table 1 Equipment List

<i>Description</i>	<i>Manufacturer</i>	<i>Model Number</i>
Solderless Breadboard	R.S.R. Electronics	MB108
Function Generator	Lab-Volt	A8771
Oscilloscope and Probes	Tektronics	DM2510
Power Supply	BK Precision	1762

Parts List:

Table 2 Parts List

<i>Quantity</i>	<i>Part</i>	<i>Value</i>
1	resistor	10 ohm - $\frac{1}{2}$ W
3	resistor	1k ohm - $\frac{1}{2}$ W
4	resistor	10k ohm - $\frac{1}{2}$ W
1	resistor	16k ohm - $\frac{1}{2}$ W
1	resistor	100k ohm - $\frac{1}{2}$ W
1	capacitor	0.01 uF
3	capacitor	0.1 uF
2	Electrolytic capacitor	10 uF
1	Dual Op. amplifier IC	LF412 or TL072CP

Procedure:

Part A: Single Amplifier Stage with GBP Constraints

- The oscilloscope is prepared by connecting both available probes to channels 1 and 2 and properly frequency compensating them while ensuring each probe is set to the 10x position.

- The power supply is configured to generate a positive 5 volt supply using only one side of the split supply, with the appropriate terminals bridged to ground, the current limit set between approximately 0.5 A and 0.6 A, proper cables used for the Vcc and ground connections, and the C.C. indicator monitored to confirm that no short circuit condition exists.
- The function generator is configured to output a 5 volt peak-to-peak sine wave at 15 kHz, which corresponds to approximately 1.06 volts peak-to-peak internally and produces an input amplitude of approximately 10 to 20 millivolts peak-to-peak at the Vin node after passing through the voltage divider, and prior to making any circuit connections both the power supply and function generator are verified to be powered off or deactivated.
- The circuit shown in the figure titled “Circuit A” is constructed on a breadboard using the provided datasheet to verify correct pin assignments, with true ground taken from the power supply ground terminal and all ground connections from the function generator and oscilloscope probes tied to this common ground, the positive 5 volt supply connected from one side of the split power supply, the Vin signal supplied from the voltage divider driven by the function generator, and the Vout signal generated by the circuit, noting that because the circuit input impedance is not close to 50 ohms the actual applied signal amplitude is approximately twice the value displayed on the function generator.
- The function generator output is connected as shown in the Circuit A diagram, channel 1 of the oscilloscope is connected to observe the function generator signal directly before the voltage divider since the reduced Vin signal is too small for reliable autoset operation, channel 2 is connected to measure Vout with proper grounding, and the instruments are powered on to observe circuit operation.
- The output waveform is verified to be approximately 800 millivolts to 1.24 volts peak-to-peak with a 2.5 volt DC offset at Vout, indicating an approximate voltage gain of 100, after which the waveform is verified by the instructor or TA and an image is saved with both the output peak-to-peak and mean voltage measurements displayed.
- The sine wave frequency is then varied in 3 kHz increments from 10 kHz to 40 kHz, with the output peak-to-peak voltage recorded at each frequency and entered into a spreadsheet program, and the measurements are verified before being presented in the lab report as a table and corresponding line graph.
- The Conclusion section of the report includes an explanation for the significant variation observed between maximum and minimum gain values across the measured frequency range.
- Upon completion of all measurements, the power supply and function generator are powered off to conclude the procedure.

Part B: Two-Stage Amplification

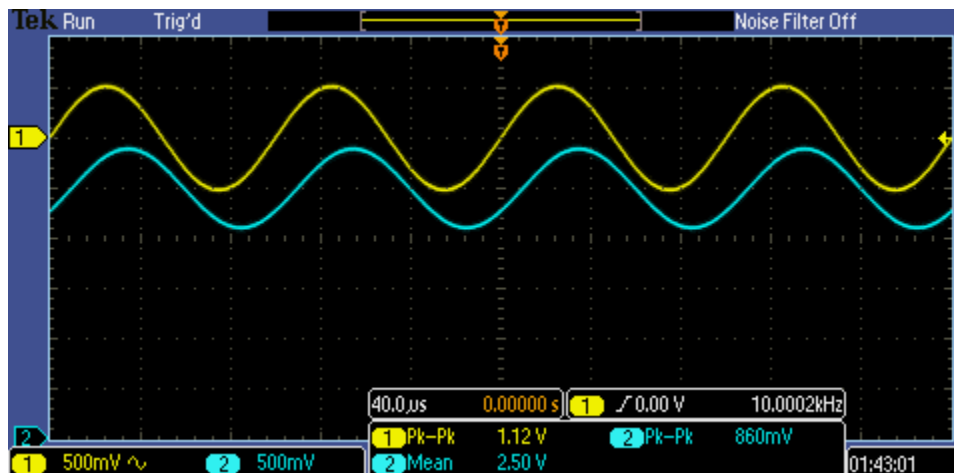
- The circuit is modified by constructing the configuration shown in the figure titled “Circuit B” on a breadboard to implement an AC-coupled amplifier stage, ensuring that the previously used 100 k Ω resistor is replaced with a 10 k Ω resistor as required for proper operation.
- Channel 1 of the oscilloscope is connected to measure the function generator output signal and channel 2 is connected to measure V_{out}, with all probe ground leads properly connected to the common circuit ground.
- The function generator is configured to output a 5 volt peak-to-peak sine wave at 15 kHz, after which the power supply and function generator are powered on to energize the circuit.
- The output waveform is observed to verify the presence of an approximately 1.28 to 1.94 volt peak-to-peak sine wave with a 2.5 volt DC offset at V_{out}, and trace averaging is enabled through the oscilloscope acquire function to obtain accurate gain measurements; if the expected waveform is not observed, the circuit connections are corrected before proceeding, and the waveform is verified by the instructor or TA prior to saving an image that includes both the output peak-to-peak and mean voltage measurements displayed on the screen.
- The sine wave frequency is then varied in 3 kHz increments from 10 kHz to 40 kHz, with the output peak-to-peak voltage recorded at each frequency and entered into a spreadsheet program for analysis.
- Using the recorded data, the gain variation is calculated and included in the Results section of the lab report, ensuring that the ratio of minimum output peak-to-peak voltage to maximum output peak-to-peak voltage over the 10 kHz to 40 kHz range exceeds 0.94, corresponding to a gain variation of less than 6 percent, with minor deviations from this target considered acceptable, and the measurements are verified by the instructor or TA.
- The recorded data are presented in the Results section as a table and corresponding line graph, along with an explanation describing why the gain variation has been reduced compared to Part A, and a calculation of the minimum-to-maximum output voltage ratio is provided using the measured data from both Part A and Part B with clear distinction between the two sets of results.
- In the Procedure section of the lab report, an explanation is included describing how the pre-lab calculations were used to select the component values implemented in Parts A

and B.

- After completing all measurements and documentation, all equipment at the station is powered off, the circuit is dismantled and components are returned to the appropriate bins without including jumper wires, cables are placed back into the basket and returned to the cart, the station is cleared, the lab verification is submitted on Canvas, the components bin and verification sheet are returned to the TA, and hands are washed upon exiting the laboratory.

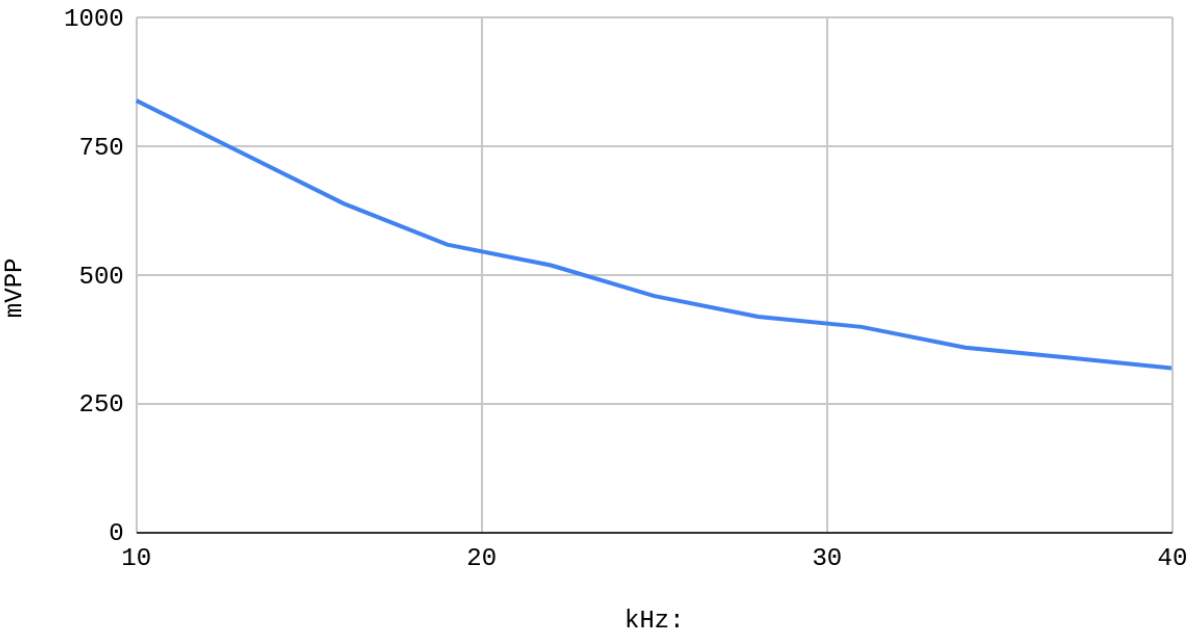
Results:

Part A:

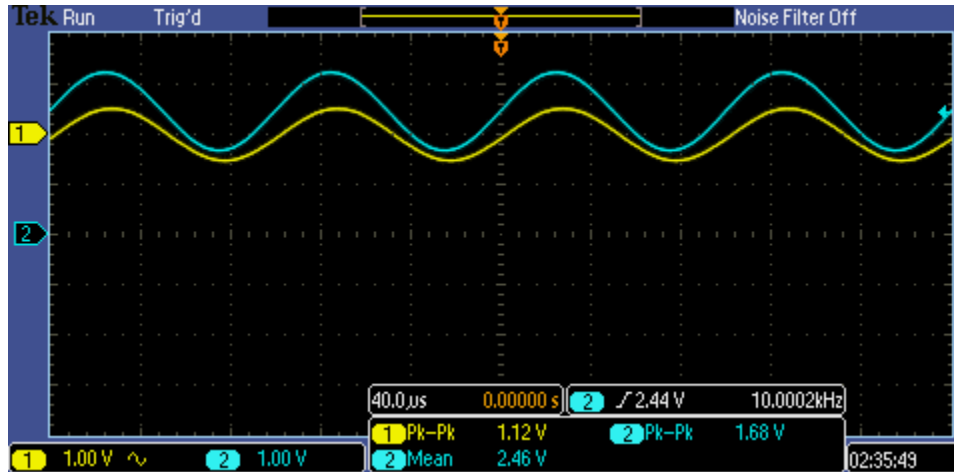


kHz:	mVPP
10	840
13	740
16	640
19	560
22	520
25	460
28	420
31	400
34	360
37	340
40	320

mVPP vs. kHz:



Part B:

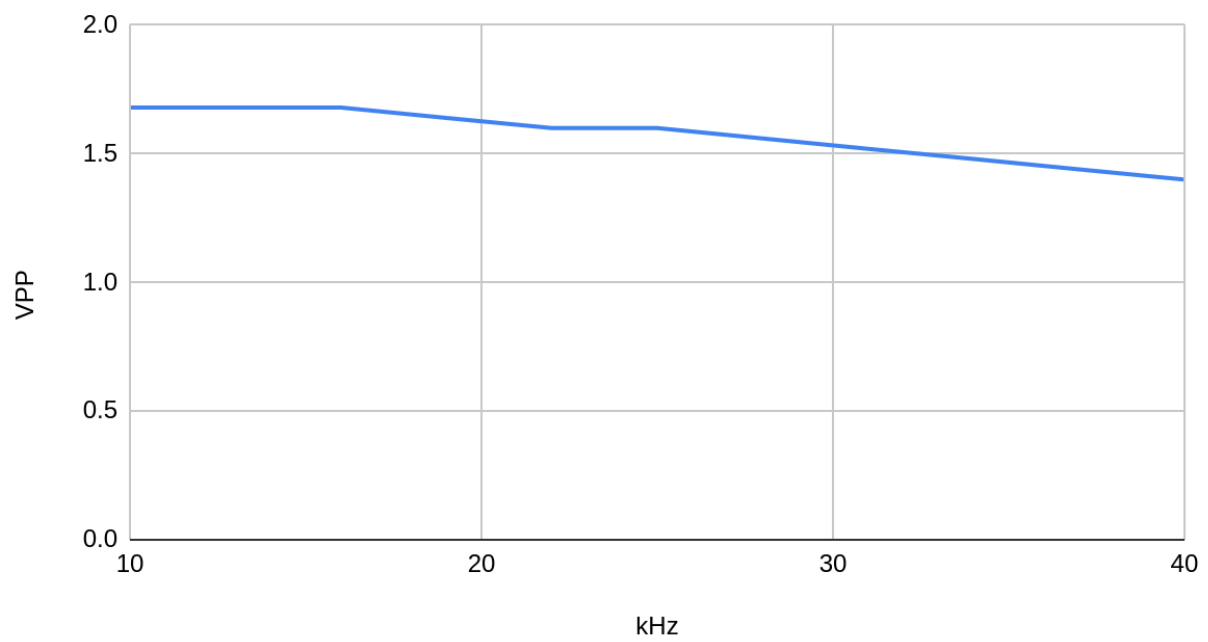


kHz	VPP
10	1.68
13	1.68
16	1.68
19	1.64
22	1.6
25	1.6
28	1.56
31	1.52
34	1.48
37	1.44
40	1.4

From the table we got a gain variation of 0.833.

Part B is cascaded through a second (the same) op amp. This decreases the sensitivity to changes in frequency (partially by adding extra steps), but also by making it so there are more factors in the final gain, resulting in frequency being a smaller part of the equation. The new results are based on the Vout of a 10k resistor at the end of the op amp as well as the 10k and 1k resistors around the new op amp.

VPP vs. kHz



Conclusion:

Part A had such a high variation between its maximum and minimum gain, because of large amounts of noise in the signal (partially) due to a high output resistance with a relatively low input resistance. The rest is due to a fairly high sensitivity to changes in frequency. A heavy dependence is also on that 100k resistor which allows for wide changes in the gain equation.